

Unit 7, Lesson 14: Multiplying and Dividing Integers with Procedural Fluency

Benchmark

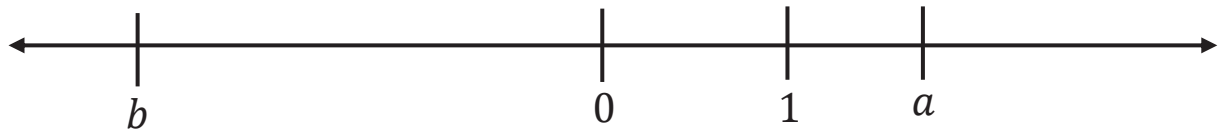
MA.6.NSO.4.2 Apply and extend previous understandings of operations with whole numbers to multiply and divide integers with procedural fluency.

Learning Target

- ☐ I can multiply and divide integers with procedural fluency.

7.14.1 Warm-Up: Thinking Algebraically

A number line is shown. The numbers 0 and 1 are marked on the line, along with two other numbers, a and b .



1. Circle one expression from those shown below that will have a negative value.
- | | | | | |
|---------|---------|---------|---------|----------|
| $a - 1$ | $a - 2$ | $a + b$ | $a - b$ | $ab + 1$ |
|---------|---------|---------|---------|----------|
2. Explain how you know the expression selected will have a negative value based on the number line.
3. Find a classmate who chose a different expression that will have a negative value.

Record their expression and explanation for why they chose it in the table. Then, have them initial that you recorded it correctly. *You are the only person who should write in the first two columns of the table.*

Classmate's Selected Expression	Classmate's Explanation for Why the Value is Negative	Initials

7.14.2 Collaboration: Multiplying Multiple Integers

1. Work with your partner to find the product of each expression.

a. $5(-9)$	b. $-6(-4)$	c. $0(-3)$
d. $(-1)(-1)(3)$	e. $4 \cdot (-2) \cdot (5)$	f. $-2(-2)(-2)(-2)$
g. $(-1) \times (-2) \times 3 \times (-2) \times (-1) \times 2$		

2. In expressions d-g, more than two factors are being multiplied.

Discuss with your partner how the rules you developed previously for multiplying two integers could be extended for multiplying more than two integers.

Write at least one rule that would apply in this situation.

7.14.3 Activity: Integer War – Multiplication Edition

Set Up:

- Remove all jokers and face cards (Kings, Queens, and Jacks) from the deck.
- With your partner, determine who is the dealer. The dealer deals the entire deck of cards to each player, face down.

Rules:

- The black cards (Spades  and Clubs ) represent positive values.
- The red cards (Diamonds  and Hearts ) represent negative values.
- Aces represent either 1 or -1 , while the other cards represent the positive or negative value indicated on the card.

How to Play:

- To begin, each player will flip over one card from their deck and place it in the center.
- The first player who figures out the product of the two integers keeps both cards and adds them to the bottom of their stack.
 - NOTE: If a player says an incorrect product, the other player responds for a chance to win the hand before the first player can change their answer.
- Repeat this process until time is called or until one player has all of the cards in the deck.

7.14.4 Guided Practice: Finding Products or Quotients

1. Find each product or quotient.

a. $10 \times (-65)$

b. -19×99

c. $-812 \div (-7)$

d. $\frac{208}{-16}$

e. $-125 \div (-5)$

f. $(-21)(-16)$

7.14.5 Your Turn!

1. Find each product or quotient.

a. -17×29

b. $41 \cdot (-19)$

c. $-184 \div 23$

d. $\frac{-175}{-5}$

e. $96 \div (-6)$

f. $(-51) \times (-31)$

g. $-256 \div (-16)$

h. $(-17) \cdot 704 \cdot (-1)$

7.14.6 Wrap-Up: Ticket Out the Door

1. Complete the table by writing a value in each blank that completes each related equation.

Multiplication Equation	...related to...	Division Equation
$(-12)(\rule{1cm}{0.4pt}) = 504$		$504 \div \rule{1cm}{0.4pt} = -12$
$\rule{1cm}{0.4pt} \times 18 = -126$		$-126 \div 18 = \rule{1cm}{0.4pt}$
$-50 \cdot \rule{1cm}{0.4pt} = -650$		$-650 \div \rule{1cm}{0.4pt} = -50$

